

Energy features

Mapping: required data, models, approaches and phases

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Abstract

This document maps 66 energy features, in 12 modules, for the map-based predictive platform, from the spatial base of the network to the analysis engine. For each feature it records the map entity it acts on, the type of output, the required data and their access level, the model, the insight or decision it produces, the client, the fit with the map requirement and the phase. The second part carries the research per module: concrete data sources, approaches (from the baseline to the recent literature), metrics, pitfalls and references.

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1 How to read the mapping

Each feature has one of six types and always acts on a map entity (requirement R3 of the specification):

Table 1
Feature types.

Type	What it is
layer	model output drawn on the map, with uncertainty, version and date
insight	finding of the analysis engine, anchored to an entity
decision	output that depends on a user constraint (plan, priority)
base	structure without a model that the others presuppose (hierarchy, boundaries)
context	third-party observed or forecast data, shown as reference (weather, satellite); not a model output of the platform
out	energy feature that does not pass the map criterion; recorded with the reason for exclusion

Table 2
Catalog columns.

Column	Values
Entity	SS: substation; CU: consumer unit
Data level	0: public data only; 1: client files; 2: scheduled connector or metering at substation and feeder; 3: continuous metering (transformer, smart metering)
Client	D distribution utility, T transmission utility, G generator, C energy trader, I end consumer
Map	●●● high: the question is spatial; ●●● medium: the map helps at portfolio or region level; ●●● low: few polygons (submarkets)
Phase	MVP V2 V3 V4 cond out . cond is the conditional track: outside the roadmap, it enters only if the target-client question of the specification is resolved in favor of generator and energy trader
Depends on	features whose output is input to this one; they define the build order

2 Overview

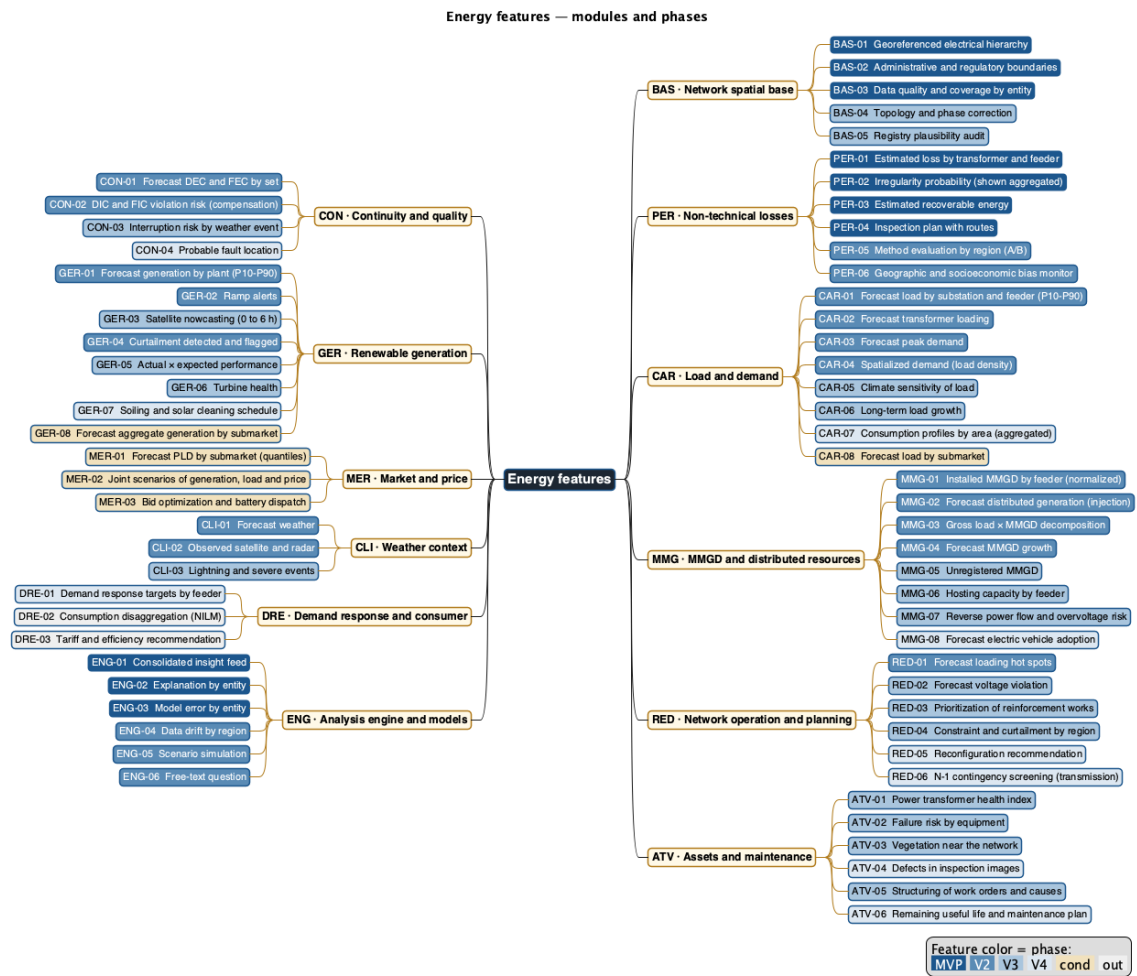


Fig. 1. Modules and features, colored by phase.

Table 3
Features by module and phase.

Module	MVP	V2	V3	V4	cond	out	Total
BAS · Network spatial base	3	–	2	–	–	–	5
PER · Non-technical losses	4	2	–	–	–	–	6
CAR · Load and demand	–	4	2	1	1	–	8
MMG · MMGD and distributed resources	–	4	3	1	–	–	8
RED · Network operation and planning	–	1	3	2	–	–	6
ATV · Assets and maintenance	–	–	4	2	–	–	6
CON · Continuity and quality	–	2	1	1	–	–	4
GER · Renewable generation	–	3	3	1	1	–	8
MER · Market and price	–	–	–	–	3	–	3
CLI · Weather context	–	2	1	–	–	–	3
DRE · Demand response and consumer	–	–	–	1	–	2	3
ENG · Analysis engine and models	3	3	–	–	–	–	6
Total	10	21	19	9	5	2	66

Table 4

How many features of each module serve each type of client.

Module	Distribution utility	Transmission utility	Generator	Energy trader	Consumer
BAS ·Network spatial base	5	2	2	1	–
PER ·Non-technical losses	6	–	–	–	–
CAR ·Load and demand	7	–	1	1	–
MMG ·MMGD and distributed resources	8	–	1	–	–
RED ·Network operation and planning	4	2	1	–	–
ATV ·Assets and maintenance	6	5	1	–	–
CON ·Continuity and quality	4	1	–	–	–
GER ·Renewable generation	–	–	7	2	–
MER ·Market and price	–	–	3	3	–
CLI ·Weather context	3	3	2	1	–
DRE ·Demand response and consumer	1	–	–	–	2
ENG ·Analysis engine and models	6	5	5	5	–

Catalog summary

- **66 features in 12 modules:** 10 in the MVP, 21 in V2, 19 in V3, 9 in V4, 5 in the conditional track and 2 out (Table 3). That is 59 committed features, 5 held as an option and 2 discarded. 47 have high map fit, 12 medium and 7 low.
- **The distribution utility concentrates the spatial value:** 50 of the 66 features serve it (Table 4). Renewable generation and market have medium or low fit, which is consistent with the client hypothesis of the specification.
- **CON-01 and CON-02 are in V2:** the BDGD carries monthly DIC and FIC per unit. CON-02 (DIC and FIC violation risk) is level 0, has high fit and measures the compensation, in reais, paid by the distribution utility. CON-01 is a prerequisite of CON-02.
- **The MVP closes the cycle:** base (BAS-01 to 03), losses (PER-01 to 04) and engine (ENG-01 to 03) cover feed → diagnosis → plan → result.

Effects of the research on the catalog

1. Data available at level 0. With public data alone, there are:

- in the BDGD: **monthly energy per feeder, technical and non-technical losses per feeder**, monthly energy, DIC and FIC per unit, and the electrical model that ANEEL runs in OpenDSS;
- at ANEEL: **every interruption since 2017 with the feeder and substation code**, cause and exclusion; DEC and FEC with limits;
- at ONS: generation by plant, half-hourly constrained-off by plant with reason and connection point, load with the MMGD share.

Therefore, 36 of the 66 features have a level 0 version, among them PER-01 (per feeder), CAR-02, RED-01, CON-01 to 03, GER-01 and GER-04; four of them (MMG-06, RED-01, RED-02 and ATV-02) depend on a level 1 or level 2 feature. **A shadow-mode pilot can be assembled before any client file is received.**

2. Limits of level 0.

- **Lag of 8 to 20 months** in the BDGD (2025 edition): it serves to assemble the network and to demonstrate, not to operate month by month.
- **The consumer unit is anonymized** (hash) in the public BDGD. Linking client data per unit requires the utility's correspondence table; through the network (feeder, transformer) the link tends to work.
- **Inconsistent units across utilities** (kWh and MWh in the same field) and, in MMGD, injection in place of generation and kW mixed with W, observed in two 2025 BDGD files. Normalization per utility and the quality layer (BAS-03) are necessary.

- The non-technical loss in the two BDGD files is consistent with **accounting apportionment**, not with measurement.

3. Three validation rules run through the catalog.

- **Store what was known at each date:** ONS and CCEE revise data after publication, and the ECMWF weather forecast stays available for only 2 to 3 days. Archiving from the first day is a requirement.
- **Validate out of space and out of event:** in weather-related interruptions, random splits inflate the performance estimate; in Essus et al. (2026), with spatial and temporal validation, many models do not beat the baseline.
- **Control multiple testing** in the engine: entities $\times$ layers $\times$ detectors require FDR and an alert budget.

4. Basis of the exploratory fraction. Revelas et al. (2025) show that selecting only the most probable targets leads to inconsistent learning; randomized selection with recorded probabilities allows unbiased estimation of prevalence, recall, calibration and the selection bias. The same fraction feeds PER-02, PER-05 and PER-06.

5. Regulatory rules are changing.

- Non-technical loss computed over the measured market since 2025, which breaks the series.
- Bid-based price formation and two PLDs planned for 2028 (Meta II).
- REN 1.098/2024 on MMGD reverse power flow.
- Portaria MME 126/2026: 2% of smart meters per year, prioritizing losses.
- Proposal for automatic Tarifa Branca.

Each one is a drift point to monitor (ENG-04).

6. Licenses to observe. BDGD under ODbL (share-alike on public derived databases: consult legal counsel); ECMWF under CC-BY with attribution on the map; *alibi-detect* under the Business Source License, with a commercial restriction.

The conditional track

Five features — MER-01 to MER-03, CAR-08 and GER-08 — left the phased roadmap for the **cond** track. They are not discarded features: they are an option on the target-client question that the specification leaves open (distribution utility, or generator and energy trader). Until that question is answered, they receive no phase.

Their removal does not affect the others: they form an isolated subtree — CAR-08 and GER-08 $\rightarrow$ MER-01 $\rightarrow$ MER-02 $\rightarrow$ MER-03 — and **nothing outside it depends on any of the five**. The external dependencies of the subtree are CLI-01 (of CAR-08) and GER-01 (of GER-08 and MER-02), features that remain in the roadmap.

Three observations weighed on the decision:

- **It is the part of the generator where the map does not help.** The five have low fit, and price by submarket occupies four polygons. If the generator track is taken, the spatial anchor is **RED-04** (constraint and curtailment by region): high fit, level 0, no dependencies, and already supported by the half-hourly constrained-off data from ONS.
- **The premise of MER-03 exists only from 2028.** Bidding only makes sense under the hybrid model of the Meta II project, and the first battery auction (Portaria MME 136/2026) pays for availability, not free arbitrage.
- **The continuity of the source is not guaranteed.** The CCEE open data portal has an application firewall; the CKAN endpoint worked on September 23, 2026.

Dependency chains

The catalog dependencies define the build order. The main ones in the phased roadmap:

- BAS-01 → PER-01, PER-02 → PER-03 → PER-04 → PER-05, PER-06 (the MVP and its evaluation);
- MMG-01 → MMG-02 → MMG-03, MMG-06, MMG-07;
- CAR-01, CAR-02 → RED-01 → RED-03, DRE-01;
- ATV-03 → CON-03, and ATV-01, ATV-02 → ATV-06.

Pending verification

1. BDGD Instruction Manual (not downloaded): official definition of energy and losses in CTMT, energy of generating units (injection or generation), unit of installed capacity.
2. Whether the set code in the BDGD matches the one in the DEC and FEC indicators.
3. Whether the distribution utility can reproduce the hash of the unit code.
4. Unit of the value in the CTR typical curves.
5. Existence of a regulatory obligation to publish hosting capacity.
6. Legal consultation on the ODbL for derived layers.

Part I

Catalog

3 Catalog by module

BAS · Network spatial base

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
BAS-01	Georeferenced electrical hierarchy	base	concession, SS, feeder, transformer, CU	BDGD (annual geodatabase, ODbL, 8 to 20 months of lag)	0	ETL, graph by connection points, unit normalization per utility	navigate, select	D	...	MVP
BAS-02	Administrative and regulatory boundaries	base	state, municipality, census tract, electrical set	IBGE (meshes), ANEEL (sets)	0	ETL	cross-reference, filter	D T G	...	MVP
BAS-03	Data quality and coverage by entity depends on: BAS-01	layer	all	all ingested sources	0	completeness and validity rules	data gap / trigger source, flag layer	D T G C	...	MVP
BAS-04	Topology and phase correction depends on: BAS-01	layer	aggregated CU, transformer	smart metering (voltage), BDGD	3	voltage series correlation, clustering	registry × measurement divergence / correct registry	D	...	V3
BAS-05	Registry plausibility audit depends on: BAS-01	insight	aggregated CU, plant	BDGD, ANEEL MMGD registry, reanalysis (NASA POWER)	0	physical ceiling (solar resource model), rules	model × registry divergence / correct registry, audit	D	...	V3

PER · Non-technical losses

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
PER-01	Estimated loss by transformer and feeder depends on: BAS-01	layer	transformer, feeder	BDGD (feeder energy, technical and non-technical losses, energy per unit), client billing, transformer metering where available	0	energy balance (feeder energy – sum of units – technical), technical loss in OpenDSS, Bayesian hierarchical apportionment among transformers	spatial anomaly, change between cycles / select targets	D	...	MVP

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
PER-02	Irregularity probability (shown aggregated) depends on: BAS-01	layer	CU aggregated by transformer	billing, registry, inspection history	1	gradient boosting with neighborhood variables, inverse inspection-propensity weighting, PU learning with propensity (SAR)	concentration / compose plan targets	D	● ● ●	MVP
PER-03	Estimated recoverable energy depends on: PER-01, PER-02	layer	transformer, feeder, plan	PER-01, PER-02, consumption	1	probability × estimated diverted energy	ranking by value / prioritize	D	● ● ●	MVP
PER-04	Inspection plan with routes depends on: PER-02, PER-03	decision	selection	PER-02, PER-03, operating base, user constraint	1	team orienteering problem (OR-Tools) with randomly drawn exploratory fraction and recorded probabilities	approve and send to the field	D	● ● ●	MVP
PER-05	Method evaluation by region (A/B) depends on: PER-04	insight	region, batch	inspection results	1	A/B test, precision at top-k	method performance / adjust inspection policy	D	● ● ●	V2
PER-06	Geographic and socioeconomic bias monitor depends on: PER-04	insight	census tract, municipality	inspections, IBGE	1	flagging rate × confirmed rate by boundary	unjustified concentration / review criteria	D	● ● ●	V2

CAR · Load and demand

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
CAR-01	Forecast load by substation and feeder (P10–P90) depends on: BAS-01, CLI-01	layer	SS, feeder	SS and feeder metering, weather forecast, calendar	2	quantile gradient boosting or TFT, hierarchical reconciliation (MinT)	trend and threshold / schedule operation	D	● ● ●	V2
CAR-02	Forecast transformer loading depends on: BAS-01	layer	transformer	BDGD (monthly energy and curve typology per unit), CTR typical curves (ANEEL), transformer metering where available	0	regulatory method (energy × typical curve in OpenDSS) calibrated on metered transformers	threshold (overload) / transfer load, replace transformer	D	● ● ●	V2
CAR-03	Forecast peak demand depends on: CAR-01	layer	feeder, transformer	measured series	2	probabilistic peak forecast by simulation with resampled weather, extreme values	trend and threshold / reinforcement, peak management	D	● ● ●	V2
CAR-04	Spatialized demand (load density) depends on: BAS-02	layer	census tract, grid	BDGD, IBGE, Open Buildings 2.5D, night lights (NASA)	0	dasymetric: consumption × buildings, population and night lights	concentration, change / plan expansion	D	● ● ●	V2
CAR-05	Climate sensitivity of load depends on: CAR-01	layer	feeder, SS	metering, reanalysis, weather forecast	2	piecewise regression with degree days, GAM	threshold in heat wave / prepare operation	D	● ● ●	V3

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
CAR-06	Long-term load growth depends on: CAR-04	layer	municipality, feeder	historical consumption, IBGE, EPE (PDE)	0	structural models with scenarios	trend / expansion plan	D	● ● ●	V3
CAR-07	Consumption profiles by area (aggregated) depends on: BAS-01	layer	transformer, census tract	smart metering or billing	3	load curve clustering	profile change / tariffs, campaigns	D	● ● ●	V4
CAR-08	Forecast load by submarket depends on: CLI-01	layer	submarket	ONS (verified and scheduled load), weather forecast	0	gradient boosting or TFT	trend / contracting	C G	● ● ●	cond

MMG · MMGD and distributed resources

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
MMG-01	Installed MMGD by feeder (normalized) depends on: BAS-01	context	feeder, transformer	ANEEL MMGD registry (monthly), BDGD	0	registry × BDGD join, unit normalization, module (DC) to inverter (AC) conversion	concentration	D	● ● ●	V2
MMG-02	Forecast distributed generation (injection) depends on: MMG-01, CLI-01	layer	feeder, SS	MMG-01, weather forecast, satellite, metering where available	0	forecast irradiance × installed capacity, calibrated	ramp, reverse power flow threshold / operate the network	D	● ● ●	V2
MMG-03	Gross load × MMGD decomposition depends on: CAR-01, MMG-02	layer	feeder, SS	CAR-01, MMG-02	2	net load = gross – MMGD, separate models	steeper duck curve / plan operation	D	● ● ●	V2
MMG-04	Forecast MMGD growth depends on: MMG-01	layer	feeder, municipality	ANEEL MMGD registry, IBGE, PDE 2035	0	hierarchical Bass with spatial covariates, EPE 4MD reference (epe4md)	trend / plan reinforcement	D	● ● ●	V2
MMG-05	Unregistered MMGD depends on: MMG-01	insight	transformer, aggregated CU	billing or smart metering, images	1	consumption profile anomaly, rooftop image detection	model × registry divergence / regularize	D	● ● ●	V3
MMG-06	Hosting capacity by feeder depends on: MMG-02, CAR-01	layer	feeder	BDGD electrical model, MMG-02, CAR-01	0	distribution power flow surrogate over scenarios	threshold (no headroom) / approve or deny connections, reinforce	D G	● ● ●	V3
MMG-07	Reverse power flow and overvoltage risk depends on: MMG-02, CAR-02	insight	feeder, transformer	MMG-02, CAR-02	0	threshold of forecast injection over minimum load	threshold / adjust tap, reinforce	D	● ● ●	V3
MMG-08	Forecast electric vehicle adoption depends on: CAR-06	layer	municipality, feeder	fleet by municipality (Senatran), IBGE	0	diffusion model with income and fleet	trend / plan reinforcement	D	● ● ●	V4

RED · Network operation and planning

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
RED-01	Forecast loading hot spots depends on: CAR-01, CAR-02	layer	feeder, transformer, SS	BDGD converted to OpenDSS, CAR-01, CAR-02	0	peak quantiles aggregated by topology, flow confirmed in OpenDSS	threshold / transfer load	D	● ● ●	V2
RED-02	Forecast voltage violation depends on: CAR-01, MMG-02	layer	feeder, bus	BDGD electrical model (bdgd2opendss), forecast load and MMGD	0	power flow surrogate (GNN), verified by solver	threshold / adjust regulation	D	● ● ●	V3
RED-03	Prioritization of reinforcement works depends on: RED-01, CAR-06, MMG-06, CON-01, CON-02	decision	feeder, SS	RED-01, CAR-06, MMG-06	1	optimization under budget: risk × cost	investment plan	D	● ● ●	V3
RED-04	Constraint and curtailment by region	layer	SS, bus	ONS constrained-off by plant (half-hourly, with reason and connection point)	0	constraint probability model by region	concentration / site new generation	G T	● ● ●	V3
RED-05	Reconfiguration recommendation depends on: RED-02	decision	feeder	electrical model, forecast load	2	GNN and optimization, verified by solver	switching	D	● ● ●	V4
RED-06	N-1 contingency screening (transmission) depends on: RED-02	layer	line, SS	client network model, forecasts	2	GNN surrogate with confirmation by full power flow	threshold / corrective actions	T	● ● ●	V4

ATV · Assets and maintenance

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
ATV-01	Power transformer health index depends on: BAS-01	layer	power transformer (SS)	oil reports (DGA), tests, loading	1	gradient boosting on dissolved gases, loading and age	change, threshold / inspect, maintain	D T	● ● ●	V3
ATV-02	Failure risk by equipment depends on: ATV-05	layer	equipment, feeder	BDGD, interruptions by event (ANEEL), client work orders, weather	0	random survival forest, count per km	concentration / preventive maintenance	D	● ● ●	V3
ATV-03	Vegetation near the network depends on: BAS-01	layer	feeder section, line	canopy height (Lang 10 m, Meta 1 m), Sentinel-2, client LiDAR	0	global canopy height maps, segmentation and height regression	concentration / tree trimming program	D T	● ● ●	V3
ATV-04	Defects in inspection images	layer	structure, equipment	drone images	1	object detection (YOLO, DETR)	concentration / work order	D T G	● ● ●	V4
ATV-05	Structuring of work orders and causes	base	equipment, feeder	free-text work orders	1	extraction with a language model	cause concentration	D T	● ● ●	V3

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
ATV-06	Remaining useful life and maintenance plan depends on: ATV-01, ATV-02	decision	equipment	ATV-01, ATV-02	1	survival and optimization under budget and crews	maintenance plan	D T	● ● ●	V4

CON · Continuity and quality

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
CON-01	Forecast DEC and FEC by set depends on: BAS-02	layer	electrical set	DEC and FEC by set (ANEEL), interruptions by event with feeder code (ANEEL, since 2017), weather	0	time series with weather	threshold against regulatory target / improvement actions	D	● ● ●	V2
CON-02	DIC and FIC violation risk (compensation) depends on: CON-01	layer	feeder, transformer (aggregated)	monthly DIC and FIC per unit in the BDGD, interruptions by event (ANEEL), TUSD	0	simulation of DIC, FIC and DMIC per unit and PRODIST compensation rules	ranking by value / prioritize actions	D	● ● ●	V2
CON-03	Interruption risk by weather event depends on: CLI-01, CLI-03, ATV-03	layer	feeder section, line	interruptions by event with cause (ANEEL), weather forecast, lightning (open GLM or BrasilDAT), ATV-03	0	forecast weather × vulnerability (vegetation, age)	threshold / pre-position crews	D T	● ● ●	V3
CON-04	Probable fault location depends on: BAS-01	layer	feeder section	recloser events, smart metering, calls	3	classification over reclosers, metering and calls	dispatch crew	D	● ● ●	V4

GER · Renewable generation

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
GER-01	Forecast generation by plant (P10–P90) depends on: CLI-01	layer	plant	generation by plant from ONS (public), plant SCADA, weather forecast	0	gradient boosting or TFT on weather forecast, zero-shot at onboarding	trend / scheduling and declaration	G C	● ● ●	V2
GER-02	Ramp alerts depends on: GER-01	insight	plant, region	GER-01	1	threshold on forecast trajectory	trend and threshold / adjust bid	G	● ● ●	V2
GER-03	Satellite nowcasting (0 to 6 h) depends on: CLI-02	layer	plant, region	GOES-19	0	U-Net on cloud index, latent diffusion (SHADECast)	change / intraday adjustment	G	● ● ●	V3
GER-04	Curtailement detected and flagged depends on: GER-01	layer	plant	ONS constrained-off by plant, SCADA	0	rules and available power model	concentration / dispute, clean training data	G	● ● ●	V2
GER-05	Actual × expected performance depends on: GER-01	insight	plant, turbine, solar block	SCADA, resource	1	normal behavior model	model × registry divergence / inspect	G	● ● ●	V3

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
GER-06	Turbine health	layer	turbine	turbine SCADA	1	normal behavior model on temperatures and vibration	change / maintenance	G	● ● ●	V3
GER-07	Soiling and solar cleaning schedule depends on: GER-05	decision	plant, solar block	SCADA, rain	1	estimated soiling loss × cleaning cost	threshold / schedule cleaning	G	● ● ●	V4
GER-08	Forecast aggregate generation by submarket depends on: GER-01	layer	submarket	GER-01, ONS	0	aggregation of forecasts	trend / commercial position	C	● ● ●	cond

MER · Market and price

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
MER-01	Forecast PLD by submarket (quantiles) depends on: CAR-08, GER-08	layer	submarket	CCEE hourly PLD (annual CSV, CC-BY), NEWAVE/DECOMP/DESSEM decks, ONS	0	gradient boosting or DESSEM emulation	trend / commercial position	C G	● ● ●	cond
MER-02	Joint scenarios of generation, load and price depends on: MER-01, CAR-08, GER-01	layer	submarket, plant	GER-01, CAR-08, MER-01	0	copulas over marginal forecasts	risk analysis	C G	● ● ●	cond
MER-03	Bid optimization and battery dispatch depends on: MER-02	decision	plant, battery	MER-02	1	stochastic optimization over scenarios	bid, dispatch	G C	● ● ●	cond

CLI · Weather context

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
CLI-01	Forecast weather	context	grid	ECMWF Open Data (CC-BY), GFS, CPTEC/INPE, INMET	0	numerical weather prediction with post-processing per entity	–	D T G C	● ● ●	V2
CLI-02	Observed satellite and radar	context	grid	GOES-19 (ABI), REDEMET and CEMADEN radar	0	precipitation nowcasting (pysteps)	–	D T G	● ● ●	V2

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
CLI-03	Lightning and severe events	context	grid	BrasilDAT (ELAT/INPE, under contract) or GOES-19 GLM (open)	0	–	–	D T	• • •	V3

DRE · Demand response and consumer

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
DRE-01	Demand response targets by feeder depends on: CAR-01, RED-01	decision	feeder	smart metering, program contracts	3	counterfactual baseline and load forecast	loading threshold / trigger program	D	• • •	V4
DRE-02	Consumption disaggregation (NILM)	out	CU	high-frequency metering	3	seq2point	–	I	• • •	out
DRE-03	Tariff and efficiency recommendation	out	CU	metering, tariffs	3	forecast and recommendation	–	I	• • •	out

ENG · Analysis engine and models

ID	Feature	Type	Entity	Data	Lvl	Model	Insight / decision	Client	Map	Phase
ENG-01	Consolidated insight feed	insight	whole hierarchy	all layers	0	LISA and Getis-Ord with FDR, change points, impact × significance score, consolidation	all detectors / follow insight	D T G C	• • •	MVP
ENG-02	Explanation by entity	layer	all	models	0	variable importance (SHAP)	–	D T G C	• • •	MVP
ENG-03	Model error by entity	layer	all	actuals	1	forecast × actual (nMAE, CRPS, precision at top-k)	model performance / recalibrate, distrust	D T G C	• • •	MVP
ENG-04	Data drift by region	insight	all	model inputs	1	two-sample tests (KS, PSI, MMD) with effect size and FDR	change / retrain	D T G C	• • •	V2
ENG-05	Scenario simulation	decision	selection	layers	0	models with altered inputs	compare alternatives	D	• • •	V2
ENG-06	Free-text question depends on: ENG-01	decision	selection	layers and insights	0	language model orchestrating queries, numbers come from the engine	–	D T G C	• • •	V2

Part II

Research by module

4 Network spatial base and non-technical losses

Inside the BDGD

Two 2025 BDGD (version V11) were downloaded and opened for this research [2]. The Instruction Manual could not be downloaded (the item on the portal responds with an error), so the semantics of some fields were inferred from the order of magnitude of the values, and this is indicated.

Access. ANEEL Open Data, ODbL license, annual update [1]. One geodatabase per distribution utility on the ANEEL ArcGIS Hub, listable through a search API and downloadable **without authentication** [2]. 2025 sizes: from about 1 MB (cooperatives) to 4 GB (Cemig-D). The ODbL has a share-alike clause for derived databases used publicly: validate with legal counsel whether the product layers fall under it (to be confirmed).

Lag. In a 2025 BDGD, the period runs from January to December 2025, with extraction in May and publication in August 2026: in that edition, publication occurred **8 to 20 months** after the reference month. The public data serve to build the network and the pilot, not for monthly operation.

Table 17
BDGD entities relevant to the catalog.

Entity	What it carries (verified in the files)
CTMT (feeder)	substation, voltages, HV transformer; monthly circuit energy (ENE_01 to 12, compatible with injected energy, to be confirmed); technical losses by segment and transformation (PERD_*); monthly non-technical losses at medium and low voltage (PNTMT, PNTBT)
UNTRMT (transformer)	rated power, iron and total losses, feeder, substation, electrical set, municipality, phases, connection date
UCBT, UCMT (units)	no geometry of their own; link to the point (PONNOT) and to the transformer with 100% fill rate in the sample; class, CNAE, load-curve typology, installed load, DG code, monthly energy, DIC and FIC ; MV also with demand. The unit code is a 64-character hash
BE, EP, PT, PNT	energy balance, pass-through energy, monthly technical and non-technical losses of the distribution utility
CONJ, ARAT, SUB	polygons of electrical set, service area and substation (the latter came empty in one of the files)
SSDMT, SSDBT	network segments, with connection points to build the graph

Pitfalls verified in the data.

- **Different units across distribution utilities:** feeder energy appears in kWh in one and in MWh in the other, and within a single table there are fields on different scales. Every loss/energy ratio must be normalized by distribution utility and by field, or the naive calculation gives “technical loss of 250%”.
- **The BDGD non-technical loss appears to be apportionment:** round values that match the aggregate table indicate accounting allocation, not measurement (to be confirmed in the Manual).
- **Implausible MMDG:** in a distribution utility in SC, annual energy over installed power reaches 4,470 kWh/kW/year, incompatible with photovoltaic generation (median 900), which is compatible with the injection and unit problems.

- **Codes:** feeder, transformer and connection points appear to be operational codes, but the consumer unit is anonymized. Linking client data *by unit* requires the distribution utility to provide the mapping; linking *through the network* tends to work (to be confirmed with a client).

ANEEL loss indicators [5–8]. In 2024, total loss was about 14% of injected energy: technical 7.4% (44.6 TWh) and non-technical 6.6% (40.2 TWh). The ten largest distribution utilities account for 74% of the non-technical loss; Light and Amazonas Energia, 34%. The cost of the actual non-technical loss was estimated by ANEEL at R\$ 10.3 billion. The regulatory limit comes from PRORET Submódulo 2.6 through benchmarking of socioeconomic complexity [9]. **Since 2025, non-technical loss has been calculated on the measured market, no longer the billed one** (Despacho 1.220/2025), which breaks the historical series. The regulatory technical loss is calculated by ANEEL by building the BDGD in OpenDSS [12].

4.1 BAS · Network spatial base

BAS-01 · Georeferenced electrical hierarchy

MVP

Data. BDGD: substation, feeder (CTMT), transformer (UNTRMT), units (UCBT, UCMT) and their points (PONNOT); segments (SSDMT, SSDBT) for connectivity. Converters to OpenDSS: *bdgd-tools* and *bdgd2dss* [11].

Approaches. Baseline: ETL with GDAL to GeoParquet or PostGIS. Recommended: graph from the connection points with validation of radiality and reachability from the feeder head; unit normalization by distribution utility; composite key (distribution utility, code); versioning by base year.

Metrics. Percentage of units linked to transformer, point and feeder; reachable segments; loops per circuit; divergence between years.

Pitfalls. Substation polygon may come empty; files of several GB; resubmissions with different dates; the BDGD is a simplified model of the network [1].

BAS-02 · Administrative and regulatory boundaries

MVP

Data. IBGE, 2022 tract mesh and aggregates [13]; the municipality in the BDGD uses the 7-digit IBGE code; electrical sets in the CONJ polygon of the BDGD and in the continuity indicators [14].

Approaches. Bridge table unit, tract, municipality, set, computed once by spatial join.

Pitfalls. 2010 and 2022 tracts incompatible; sets change between years; matching the BDGD set code with that of the indicators (to be confirmed).

BAS-03 · Data quality and coverage by entity

MVP

Typical BDGD problems. Disconnected sections, inconsistent buses, wrong links, incorrect phase, inadequate impedances, misallocated load; a single parameterization error changed the calculated technical loss by up to 1.45% [12].

Approaches. Completeness and validity rules by field (Great Expectations, pandera) and a **score by entity** with physical rules (total loss greater than iron loss; yield below the cap; loss smaller than energy), consistency rules (sum of the units smaller than the feeder energy), unit detection by order of magnitude and missing links. The score becomes a layer and qualifies the others.

Pitfalls. Rules calibrated on one distribution utility fail on another because of the units; the quality layer can be read as an accusation against the client.

BAS-04 · Topology and phase correction

V3

Data. Voltage series from smart meters (level 3) and the BDGD. Portaria Normativa MME 126/2026 mandates additional deployment in 2% of units per year for 24 months starting in March 2026, prioritizing reduction of non-technical loss [17].

Approaches. Baseline: correlation of the voltage series. Recommended: clustering on voltage correlation with a transformer capacity constraint, generating candidates for field validation (Short 2013 [19]; Hoogsteyn et al. 2022: voltage series outperform power series [20]).

Pitfalls. Sparse coverage per transformer; synchronization between meters; MMGD alters the voltage profile.

BAS-05 · Registry plausibility audit

V3

Data. BDGD generating units (power, energy, DG code); ANEEL MMGD registry [15]; NASA POWER [16].

Approaches. Baseline: annual yield cap by municipality from irradiation. Recommended: expected monthly range from a physical model (pvlib), compared with the recorded injection, which is a lower bound on generation; cross-check the BDGD power against the registry to detect W and kW. Real example: 4,470 kWh/kW/year in SC.

Pitfalls. Injection depends on self-consumption: the test works as a cap, not as a floor; remote compensation; missing DG code.

4.2 PER · Non-technical losses

PER-01 · Estimated loss by transformer and feeder

MVP

Data. Public: feeder energy, technical and non-technical losses from CTMT, energy by unit, iron and total losses of the transformers. Client: monthly billing (the BDGD is annual and lagged) and transformer metering when available.

Approaches. **Baseline by feeder, with public data only:** feeder energy minus the sum of the units (LV, MV, public lighting) minus injected generation, compared with the BDGD losses. By transformer without metering: technical loss from iron plus copper proportional to the square of the load; the non-technical loss is left as the residual. Recommended: power flow in OpenDSS for the technical loss (as ANEEL does) and balance for the total, with a Bayesian hierarchical model that splits the feeder residual among the transformers using the attributes from PER-02. Recent literature: technical loss without power flow from sparse voltage [41]; estimation of rates by region in a Brazilian city [43].

Metrics. Error against metered transformers; stability between cycles; agreement with the regulatory total.

Pitfalls. Mixed units; BDGD non-technical loss appears to be apportionment; billing cycle different from the calendar month; estimated public lighting; energy injected by DG enters the balance.

PER-02 · Irregularity probability

MVP

Data. Client: inspections with outcome (positive and negative), billing, meter replacements, disconnections. BDGD: class, CNAE, installed load, DIC and FIC, DG code. Context: census tract, transformer loss (PER-01), neighborhood.

Approaches. Baseline: rules and gradient boosting on the inspections. Recommended: boosting with **neighborhood variables** [23], selection bias correction by inverse inspection propensity weighting, calibration on the exploratory fraction and PU learning.

What the literature says. Glauner et al. showed covariate shift in a Brazilian dataset of 3.6 million clients and 820 thousand inspections: inspections concentrated in neighborhoods and classes make the models biased [22,24]. Coma-Puig and Carmona propose measuring recovered energy and explaining [25,26]. Revelas, Boldea and Werker (2025) show that selecting only the most probable leads to inconsistent learning and that randomized selection allows updating the model as if the sample were random [35]: this is the basis adopted for the exploratory fraction. In PU learning [31–33], the assumption that the labeled positives were chosen at random **does not hold** in losses, because inspections are targeted: use variants with propensity dependent on the variables (SAR-PU) [34].

Brazilian cases. Light: satellite and BDGD for illegal connections [37], ANEEL R&D with GESEL/UFRJ on areas with severe operating restrictions [39]; Optimum-Path Forest (UNESP, Unicamp) [28]; survey of 28 experts [36]; irrigators in RS [44].

Metrics. Precision at top-k with k equal to the inspection capacity; energy-weighted recall; calibration **measured on the exploratory fraction**; AUC only as a diagnostic.

Pitfalls. Metrics inflated by oversampling and leakage; inspection negative is noisy; drift after meter replacement; LGPD.

PER-03 · Estimated recoverable energy

MVP

Approaches. Baseline: probability $\times$ estimated diverted energy (consumption before the drop or median of the peers). Recommended: two stages, probability $\times$ severity, with severity regressed on the confirmed positives, optimizing net return (recovered minus inspection cost), as in Massafferro, Di Martino and Fernández (UTE, Uruguay) [27].

Pitfalls. Severity is observed only in the positives (another bias); accounting recovery is not sustained recovery; double counting with the transformer balance.

PER-04 · Inspection plan with routes

MVP

Approaches. Baseline: top by expected value and nearest-neighbor route. Recommended: **team orienteering problem** (routing with prizes) maximizing expected value under each team's work shift, solved with OR-Tools; **exploratory fraction drawn at random by stratum with recorded probabilities** (5 to 15%) [35], used later as weights in PER-02, PER-05 and PER-06. No specific literature on routing for losses was found.

Metrics. Value captured over the optimum; inspections per team and day; cost per productive inspection; percentage of the plan executed.

Pitfalls. Safety criteria take precedence over the random draw; teams that do not follow the random draw; grouping targets by route reintroduces geographic bias.

PER-05 · Method evaluation by region (A/B)

V2

Approaches. Randomization **by cluster** (transformer, neighborhood or team) to avoid contamination; random arm as the prevalence reference; Horvitz–Thompson estimators; effect measured in recovered energy and in feeder loss over time (difference in differences).

Pitfalls. Low statistical power with low prevalence; deterrence effect among neighbors; change of the regulatory metric in 2025.

PER-06 · Geographic and socioeconomic bias monitor

V2

Approaches. Compare, by breakdown, the flagging rate with the confirmation rate **estimated on the random fraction**; calibration and true positive rate by group; proportion tests with correction for multiple comparisons; Bayesian smoothing in small tracts. No specific work on fairness in losses in Brazil was found. Regulation recognizes socioeconomic complexity as a determinant, so different prevalences are expected [6].

Pitfalls. Without a random arm, bias cannot be separated from real prevalence; re-identification in fine breakdowns; using income as a variable and auditing the same variable.

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5 Load, demand and MMGD

Shared sources

Table 18
Sources shared by the CAR and MMG modules.

Source	What it provides	Pitfalls
BDGD (ANEEL) [1,2]	annual geodatabase by distribution utility, reference date December 31, published about 8 months after the reference date in the 2025 edition. In the unit tables: feeder, transformer, substation, municipality, class, CNAE, curve typology, monthly energy and demand, monthly DIC and FIC , DG code and coordinates	the open CSVs only carry legal entities ; residential requires the full geodatabase. DG energy as injection and power in kW and W mixed: observed empirically in a subset, not confirmed in the specification
MMGD registry (ANEEL) [4,5]	monthly; municipality, substation and approximate coordinates, modality (local, remote, shared), class, source, power in kW	power is the sum of the modules (DC), not of the inverter; there is no feeder, only substation
CTR typical curves (ANEEL) [6]	typical consumer and typical network from the measurement campaigns of the tariff reviews, by day type, in 15-minute intervals	unit of the value not stated; each distribution utility has one campaign year; curves predate the MMGD expansion
ONS, verified and scheduled load [7,8]	half-hourly by load area, with components, including the share served by MMGD	breaks in 2021 (global load) and 2023 (estimated MMGD enters the load) [9]; ONS advises not using years before 2022 [10]
IBGE, 2022 Census [11,12]	aggregates and census tract mesh	2010 and 2022 tracts are not comparable without harmonization
Nighttime lights (NASA Black Marble) [13,14]	500 m, daily, monthly and annual	public lighting is not consumption; urban saturation
Google Open Buildings 2.5D Temporal [15]	building presence, count and height, annual from 2016 to 2023, covers Brazil	raster, no polygon; ends in 2023
Senatran, fleet [16,17]	monthly by municipality and fuel since 2003	fleet registered at the owner's domicile; rental companies concentrate registrations
EPE [18–21]	PDE 2035 (MMGD, electromobility, demand); the 4MD model and the R package <i>epe4md</i>	national or by distribution utility; the disaggregation is the user's

5.1 CAR · Load and demand

CAR-01 · Forecast load by substation and feederV2

Data. Substation and feeder metering at 15 minutes (client); weather forecast and reanalysis; calendar; BDGD topology; MMGD by feeder (MMG-01).

Approaches. Baseline: similar profile and regression with temperature and calendar (Tao Vanilla style) [22]. Recommended: one global quantile boosting model for all feeders (P10, P50, P90), followed by hierarchical reconciliation transformer, feeder, substation, area with MinT [23]; probabilistic version as in Ben Taieb et al. with smart meters [24]. Recent literature: TFT [26] and foundation models [27–30]; **Chronos-2** accepts native covariates, needed for weather [28]; use as challenger and for feeders with little history [31,32].

Metrics. Pinball and CRPS [33]; P10–P90 coverage (expected 80%); WAPE of the P50 by horizon; zero-sum residual after reconciling.

Pitfalls. Switching operations and transfers create false steps (requires a switching log); what is metered is net load; MinT with thousands of series needs covariance with shrinkage.

CAR-02 · Forecast transformer loading

V2

Data. BDGD: monthly energy and curve typology by unit, link to the transformer and rated power [1]; CTR curves [6]; weather; sample metering of the client's transformers, when available.

Approaches. Baseline: the **regulatory method** for technical losses, which ANEEL has run in OpenDSS since 2014 from the BDGD, with monthly energy $\times$ typical curves in 24 levels [34,35]. Recommended: the baseline plus a calibration factor learned on metered transformers (boosting with class composition, number of units, weather and MMGD), to reduce the underestimation of diversity; output as peak distribution and probability of exceeding 100% or 120%. Conversion to OpenDSS: *bdgd2dss* [36].

Metrics. Peak error on metered transformers; AUC-PR of overload; Brier.

Pitfalls. CTR curves old and without MMGD [3]; wrong unit to transformer link in the BDGD; rated power is not thermal capacity.

CAR-03 · Forecast peak demand

V2

Approaches. Baseline: historical maximum $\times$ growth. Recommended: peak density forecast by simulation, feeding the hourly model with many years of resampled weather and extracting the 10, 50 and 90% exceedance probabilities (Hyndman and Fan [37]; Hong, Wilson and Xie [38]). Recent literature: extreme values (GEV, GPD) on residuals.

Metrics. Pinball at the high quantiles; error of the peak value and time; hit rate of "peak day".

Pitfalls. With MMGD the net peak shifts to the early evening; few extremes in the history.

CAR-04 · Spatialized demand

V2

Data. Consumption by unit with coordinates (full BDGD or client); 2022 Census tracts; Open Buildings 2.5D; Black Marble.

Approaches. Baseline: direct aggregation of units into a hexagonal grid or tract, where there are coordinates. Recommended: dasymetric in WorldPop style, with a random forest that predicts consumption density from built area and volume, population, nighttime light and land use, redistributing the known total with a sum constraint [39,40]. Recent literature: downscaling with deep learning at about 130 m [41].

Metrics. Spatial cross-validation (municipality left out) against the actual aggregation; exact mass conservation.

Pitfalls. Nighttime light measures lighting; **LGPD**: aggregate with a minimum number of units per cell.

CAR-05 · Climate sensitivity of load

V3

Approaches. Baseline: piecewise regression with degree days; the slope above the balance point (MW/°C) is the indicator. Recommended: GAM with temperature spline, hour $\times$ temperature interaction, lagged effects from 24 to 72 h and Bayesian hierarchy across feeders. State of the art: distributional regression [42]. Context: the IEA estimates +3.6 GW of demand per +1 °C in Brazil [43,44].

Pitfalls. Estimate on gross load (a hot, sunny day reduces the net load); air-conditioning penetration changes the sensitivity over the years.

CAR-06 · Long-term load growth

V3

Approaches. Baseline: PDE rate applied top-down [45]. Recommended: structural model by class with scenarios, reconciled with the PDE at the top, and spatial allocation by built-area growth; large point loads (data centers, industry) as events.

Metrics. Rolling-origin backtest over 3 to 10 years; spatial allocation error; scenario coverage.

CAR-07 · Consumption profiles by area

V4

Approaches. Without smart metering, synthetic profile from the class composition and CTR curves; with metering, clustering (k-medoids with DTW, GMM) evaluated by Chicco's indices [46].

Pitfalls. A synthetic profile only reproduces the existing typologies; LGPD in small aggregations.

CAR-08 · Forecast load by submarket

cond

Approaches. Baseline: the ONS scheduled load [8]. Recommended: boosting or TFT by subsystem with load-weighted weather and the MMGD component modeled separately [7]; reconciliation area, subsystem, SIN.

Pitfalls. Breaks of 2021 and 2023 [9]; retroactive revisions.

5.2 MMG · MMGD and distributed resources

MMG-01 · Installed MMGD by feeder

V2

Approach. (1) Join registry and BDGD by the DG code; on failure, spatial join within the same substation. (2) Normalize units by comparing with the registry power (a ratio close to 1,000 indicates W). (3) For the network what counts is the plant location, not the unit that receives the credit (remote, shared). (4) Convert modules (DC) to inverter (AC) with the sizing factor used by EPE [20].

Metrics. Join rate; power divergence between sources by feeder.

Pitfalls. BDGD with 8 to 20 months of lag and monthly registry: use the registry for counts and the BDGD for topology; approximate coordinates lead to the wrong feeder in dense areas.

MMG-02 · Forecast distributed generation

V2

Approaches. Baseline: installed power $\times$ irradiance $\times$ PR (0.75 local and 0.80 remote, 4MD assumptions) [20]. Recommended: physical model with pvlib and upscaling by reference systems, with quantile post-processing calibrated on the feeder metering. ONS estimates MMGD with the ANEEL capacity, INPE irradiation and its own capacity factor [9].

Generation and injection. Generation is not injection: injection is generation minus simultaneous self-consumption, which requires a model of the unit's load.

Pitfalls. Unknown orientation; unregistered capacity; inverter limiting.

MMG-03 · Gross load $\times$ MMGD decomposition

V2

Approaches. Baseline: add the estimated generation to the net load. Recommended: joint estimation, net = load(weather, calendar) $- k \times$ PV proxy(irradiance), with k (effective power) by regression or Kalman filter; the estimated k audits the registry and feeds MMG-05. References: SunDance [48], unsupervised disaggregation [49], PNNL [50], game theory [51], solar exemplars [52].

Pitfalls. Collinearity between irradiance, temperature and air conditioning; k changes over time.

MMG-04 · Forecast MMGD growth

V2

Reference model. EPE's 4MD: Bass with dynamic potential market, 52 distribution utilities, segments residential local, other LV, HV and remote; potential market by income; adopting fraction dependent on payback; available as the R package *epe4md* [20,21]. PDE 2035: 61.4 to 97.8 GW of MMGD in 2035 [18].

Approaches. Baseline: *epe4md* by distribution utility, allocated by current share. Recommended: hierarchical Bass (distribution utility, municipality, feeder) with spatial covariates; in Minas Gerais, wage, higher education and urbanization have the largest effects and there is spatial spillover [53,54]. Recent literature: agents and system dynamics [55]; DeepSolar [56].

Pitfalls. Regulatory shocks (Lei 14.300/2022, REN 1.098/2024) that the classic Bass does not capture, as EPE acknowledges [20].

MMG-05 · Unregistered MMGD

V3

Approaches. Baseline: persistent drop in consumption without a DG code and inverted seasonality, with change point and correlation with irradiance (Zhang and Grijalva 2016) [57]. Recommended: this screening plus confirmation by computer vision on the candidates, with Brazilian precedents [58,59]; at the feeder, the MMG-03 residual. Recent literature: Bayesian fusion of registry and remote sensing [60]; attention to the low reliability of detectors outside the training scale and region [61].

Pitfalls. Consumption drop from other causes; an indication is not proof; aggregated display.

MMG-06 · Hosting capacity by feeder

V3

Context. REN 1.098/2024 created cases of exemption from the reverse power flow analysis and an order of solutions before denying access (power reduction, time restriction, change of connection point, works) [63,64].

Approaches. Deterministic: increase PV in the worst case until a violation. Stochastic: Monte Carlo over location and size; Torquato et al. (Unicamp, 2018), on 50,000 real low-voltage networks, identified overvoltage as the most restrictive limit and approximately lognormal hosting [65]. Time series over 8,760 h [66]. **Recommended for scale:** OpenDSS on a stratified sample of feeders and a surrogate that predicts hosting from feeder attributes [67]. Recent literature: PINN [68].

Metrics. Surrogate error against OpenDSS and hit rate of the limiting factor; agreement with access assessments.

Pitfalls. Inconsistent phases and impedances in the BDGD; unknown primary tap; publishing “headroom” has regulatory and commercial effect.

MMG-07 · Reverse power flow and overvoltage risk

V3

Approaches. Baseline: P90 of injection over P10 of daytime load, by transformer and feeder. Recommended: joint sampling of load and PV with a copula and voltage sensitivity from the MMG-06 surrogate, calibrated with the client’s complaints and measurements.

Pitfalls. Minimum load of an unmetered transformer comes from outdated curves; underreported events.

MMG-08 · Forecast electric vehicle adoption

V4

Data. Senatran fleet [16]; PDE 2035: electric vehicles at 23% of light-vehicle registrations in 2035 and demand of 0.6 to 7.8 TWh between 2025 and 2035 [19].

Approaches. Hierarchical Bass by municipality constrained to the PDE total, separating plug-in electric vehicles from hybrids; load by vehicles $\times$ km $\times$ kWh/km $\times$ charging profile, allocated by CAR-04.

Pitfalls. Fleet at the owner’s domicile; little measured charging data in Brazil.

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6 Network operation and continuity

Shared sources

Table 19
Sources shared by the RED and CON modules.

Source	What it provides	Pitfalls
ANEEL, Indicadores Coletivos de Continuidade [1]	measured DEC and FEC and limits by set and month, components, compensation paid, set attributes; since 2000	limits change at each tariff review; sets are reconfigured
ANEEL, Interrupções nas Redes de Distribuição [2]	every interruption since 2017, with BDGD feeder and substation code , set, start and end, cause, reason for exclusion (expurgo), consumers affected; ODbL	cause reported by the distribution utility; does not include permit-holder utilities
ANEEL, BDGD [6] and conversion to OpenDSS [7,8]	ANEEL computes technical losses by building the BDGD network in OpenDSS; <i>bdgd2openss</i> (MIT) replicates the methodology; alternatives validated in an undergraduate thesis (4.1% error in total active power)	static model; does not model the consumer's net generation , which limits reverse power flow studies
ONS, constrained-off of wind and photovoltaic plants by plant [10]	half-hourly since 2021, with generation, limited generation, reference, reason for the constraint (REL, CNF, ENE, PAR), origin (local or systemic), connection point; CC-BY	reference generation is an estimate; data are reprocessed
ONS, transmission lines and availability [12]	registry of lines above 230 kV; availability of transmission functions by month	registry without history
EPE, power flow cases [13]	ANAREDE cases from the PDE and the PAR/PEL	CEPEL format; laborious conversion
Lightning [16,17]	RINDAT and BrasilDAT (access by agreement or contract); GLM from GOES, free	GLM does not separate cloud-to-ground from intracloud

PRODIST Module 8 rules (REN 956/2021) used as parameters [4] .

- Voltage from 2.3 to 69 kV: adequate between 0.93 and 1.05 pu; precarious between 0.90 and 0.93; critical below 0.90 or above 1.05. Low voltage 220/127 V: adequate from 202 to 231 V and from 117 to 133 V.
- DRP and DRC over 1,008 10-minute readings, limits of 3% and 0.5%, with compensation proportional to the excess.
- Compensation for DIC, FIC and DMIC computed with the VRC (TUSD fio B) and factors by voltage level; cap of 18 times the VRC; in a simultaneous violation only the largest is paid; DICRI separately.
- Exclusions from DEC and FEC include interruptions in an emergency situation and Critical Day, defined as the day on which the set's emergency occurrences exceed the mean plus three standard deviations of the previous 24 months [5].

Check the version of the module in force before coding the rules [4].

6.1 RED · Network operation and planning

RED-01 · Forecast loading hot spots

V2

Data. Level 0: transformer and substation power rating, monthly energy by unit and typical curves from the BDGD, converted to OpenDSS [6,7]. Level 1: client billing; level 3: transformer metering. Admissible overload by the hot-spot thermal model (IEEE C57.91) [42].

Approaches. Baseline: peak estimated by energy $\times$ typical curve over rated power, projected with load growth. Recommended: peak quantiles by transformer and feeder, aggregated by the topology, with output as probability of exceeding the limit and flow confirmed in OpenDSS. Recent literature: overload alarm classifiers with smart metering [38].

Metrics. Precision and recall at top-k against replacements due to burnout or metering; Brier and calibration of the overload probability; pinball of the peak.

Pitfalls. Monthly energy $\times$ typical curve smooths the peak; the overload label only exists where there has already been a problem; outdated nameplate; load transfers outside the static model.

RED-02 · Forecast voltage violation

V3

Data. BDGD electrical model through *bdgd2opendss* [7,8]; forecast load and MMGD; Module 8 limits [4]; labels generated by time-series simulation in OpenDSS over scenarios; validation with complaints and sampled DRP and DRC measurements from the client.

Approaches. Baseline: full OpenDSS by feeder in the peak scenario and the maximum-MMGD with minimum-load scenario, feasible in batch. Recommended: GNN surrogate trained on the simulations of the client's network, ranking buses and sections, with **every alert confirmed by OpenDSS**. Recent literature: PowerFlowNet [26]; PowerFlowMultiNet for unbalanced three-phase networks, about 100 times faster [27]; physical biases for voltage in distribution [28]; PFA generalization benchmark [29].

Metrics. Voltage error in pu; **recall of violations** above precision; fraction of alerts confirmed; performance on feeders not seen in training.

Pitfalls. Without modeling of net generation, overvoltage from MMGD needs its own model [7]; regulator and capacitor settings may be missing; GNN degrades outside the training topologies [29].

RED-03 · Prioritization of reinforcement works

V3

Data. Risks from RED-01, CAR-06 and MMG-06; the client's work costs; taxonomy of the Plano de Desenvolvimento da Distribuição (expansion, improvement, renewal) [15]; continuity benefits in reais (CON-01, CON-02).

Approaches. Baseline: greedy ranking by risk avoided per real. Recommended: multi-year integer programming (knapsack) with dependencies, mutually exclusive alternatives, annual limit and load and MMGD scenarios, with the benefit of each work evaluated in OpenDSS. Recent literature: two-stage stochastic or robust planning [39].

Metrics. Risk reduced per real; regret in backtest; stability of the ranking across runs.

Pitfalls. Benefit depends on uncertain forecasts (MMGD); regulatory recognition of the investment not modeled (to be confirmed).

RED-04 · Constraint and curtailment by region

V3

Data. Constrained-off by plant from the ONS, half-hourly, with reason, origin and connection point [10], measured by routine RO-AO.BR.13 [11]. For the map, georeference the connection point through the registry of lines and substations [12].

Approaches. Baseline: aggregation of unrealized generation by connection point, reason and origin, with monthly trend. Recommended: model of constraint probability and expected energy by region, separating the energy constraint (systemic) from the electrical ones (local). Recent literature: reduced network (PTDF) over the EPE cases, calibrated with the history, to estimate the marginal curtailment of new generation (to be confirmed).

Pitfalls. Reference generation is an ONS estimate; only the REL reason goes on to reimbursement at the CCEE; does not cover MMGD.

RED-05 · Reconfiguration recommendation

V4

Data. OpenDSS model of the BDGD with switches and normal state; the client's remote-controlled switches; forecast load; radiality, voltage, thermal limit and protection constraints.

Approaches. Baseline: branch exchange of Baran and Wu (1989) [33]. Recommended: mixed-integer conic optimization over DistFlow generates a few configurations, each one verified in OpenDSS and presented as a suggestion. Recent literature: reinforcement learning over graphs for restoration in real time, tested on IEEE networks [34].

Pitfalls. Protection coordination outside the model; switch state in the BDGD may not reflect operation; gap between test and real networks. Shadow mode only.

RED-06 · N-1 contingency screening (transmission)

V4

Data. Model of the client's network or ANAREDE cases from the EPE [13]; unavailability of transmission functions [12]; financial impact through the Parcela Variável por Indisponibilidade (REN 729/2016) [14].

Approaches. Baseline: distribution factors (PTDF, LODF) and full AC on the most severe ones. Recommended: GNN calibrated for recall close to 100%, with full AC on everything above a low threshold. Recent literature: PINN with GNN for N-1 to N-3, about 400 times faster, tested up to 118 buses [31]; input-convex networks with a reliability guarantee [32].

Pitfalls. Since a false negative is unacceptable, the model only orders the cases for the full AC. The literature goes up to 118 buses, much smaller than the SIN; the official security assessment belongs to the ONS.

6.2 CON · Continuity and quality

CON-01 · Forecast DEC and FEC by set

V2

Data. DEC and FEC with limits [1]; interruptions by event linked to the feeder since 2017 [2]; emergency occurrences [3]; weather (INMET, reanalysis) and lightning.

Approaches. Baseline: seasonal naive and ETS by set. Recommended: monthly global model by set over the regulated components, with the exclusions modeled separately; output as quantiles and probability of exceeding the annual limit. Recent literature: frequency $\times$ duration by feeder in Monte Carlo over weather scenarios, reconciled up to the set (to be confirmed). In Brazil: neural networks for indicators [41] and ML in the definition of set limits [40].

Metrics. MAE and pinball of DEC and FEC; Brier of the violation probability; hit rate of the ranking of sets at risk.

Pitfalls. The Critical Day uses a 24-month moving threshold and makes the exclusion dependent on the weather [5]; reconfiguration of sets breaks series.

CON-02 · DIC and FIC violation risk (compensation)

V2

Data. Module 8 formulas and factors [4]; compensation aggregated by set in the open data [1]. **Monthly DIC and FIC by unit are in the BDGD**, in the UCBT and UCMT tables, verified in the files: sizing the exposure is **level 0**. Caveat: the BDGD has a lag of 8 to 20 months and the unit is anonymized by hash, so the public data serve to size and validate the method, not to track the current quarter — that is level 1, with the distribution utility's code mapping.

Approaches. Baseline: historical violation rate and mean compensation by feeder. Recommended: simulate DIC, FIC and DMIC by unit from an event model by section, apply the compensation rules (maximum, cap) and aggregate the expected value by feeder or transformer.

Metrics. Compensation error in reais by feeder and quarter; precision at top-k of the ranking.

Pitfalls. Monthly, quarterly and annual limits at the same time; nonlinear rules; LGPD on the data by unit.

CON-03 · Interruption risk by weather event	V3
Data. Labels: interruptions with cause and exclusion, and emergency events [2]. Weather forecast (CLI-01), stroke density from CLI-03 [16,17], vulnerability (vegetation from ATV-03, network age and type from the BDGD). Approaches. Baseline: count regression (Poisson or negative binomial) by cell and day over gust, rain and lightning. Recommended: the pattern of the UConn model, in operation since 2015: tree ensemble by event on a 2 km grid, with weather, vegetation and infrastructure variables and modules by weather type [19–21]. Recent literature: storms as tracked objects [20]; two stages with open data [24]; GNN with tree risk from LiDAR [25]. Metrics. Error of the total interruptions by event; spatial POD, FAR and CSI; skill by lead time. Leave-one-event-out and leave-one-region-out validation. Pitfalls. Random splits inflate the performance estimate: with spatial and temporal validation, many models do not beat the baseline [23]. Weather forecast error propagates. Transfer of models from temperate climates to tropical convection is not proven.	
CON-04 · Probable fault location	V4
Data. Recloser and relay events (sequence of events, current, oscillography), last signal from the smart meters, customer calls, OpenDSS topology. Requires level 3 for real-time use. Approaches. Baseline: topological inference (common ancestor of the units without power, downstream of the device that operated) combined with the impedance method in OpenDSS. Recommended: physical candidates ranked by a classifier with cause priors (vegetation, lightning, history). Recent literature: spatio-temporal GNN on partially observed networks [36]; digital twin with low-voltage meters [37]. Metrics. Section hit rate at top-1 and top-3; error in km; reduction in kilometers patrolled. Pitfalls. Scarce real labels; multiple faults in a storm; DG distorts the impedance method.	

CON module diagrams

The module has two rhythms: the **regulatory** one, monthly, over sets and feeders (CON-01, CON-02), and the **event** one, from hours to days, triggered by each weather forecast run (CON-03). The diagrams show on which entities each layer is drawn, who uses what, the states of the two central objects (the weather event and the set in the regulatory year) and the flow that links data, models and decisions.

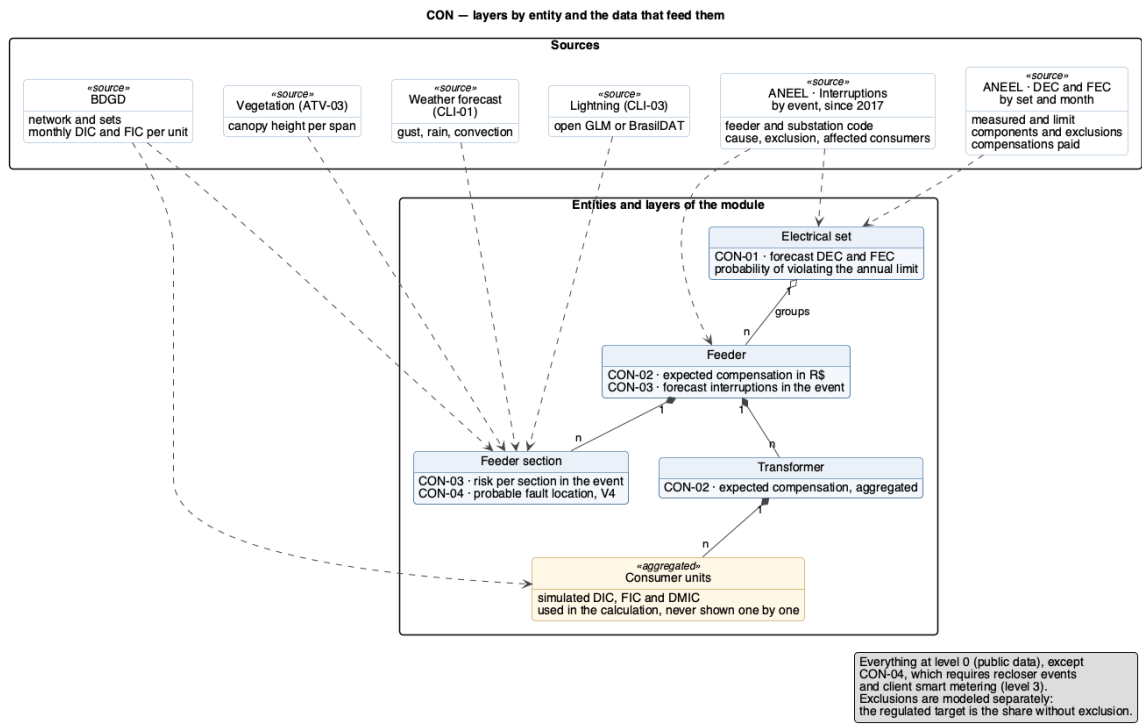


Fig. 2. CON: layers by entity and the data that feed them. Level 0, except CON-04; CON-02 requires level 1 for the current quarter.

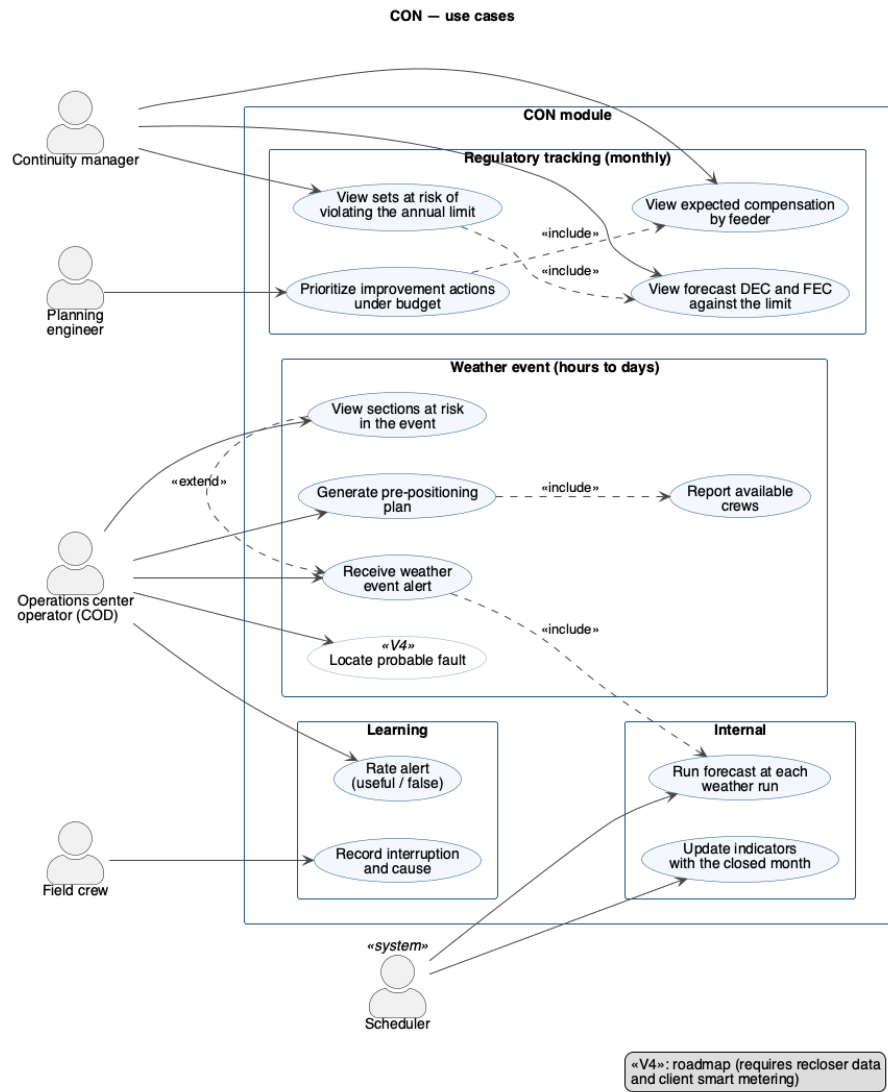


Fig. 3. CON: use cases, separated by the module's two rhythms.

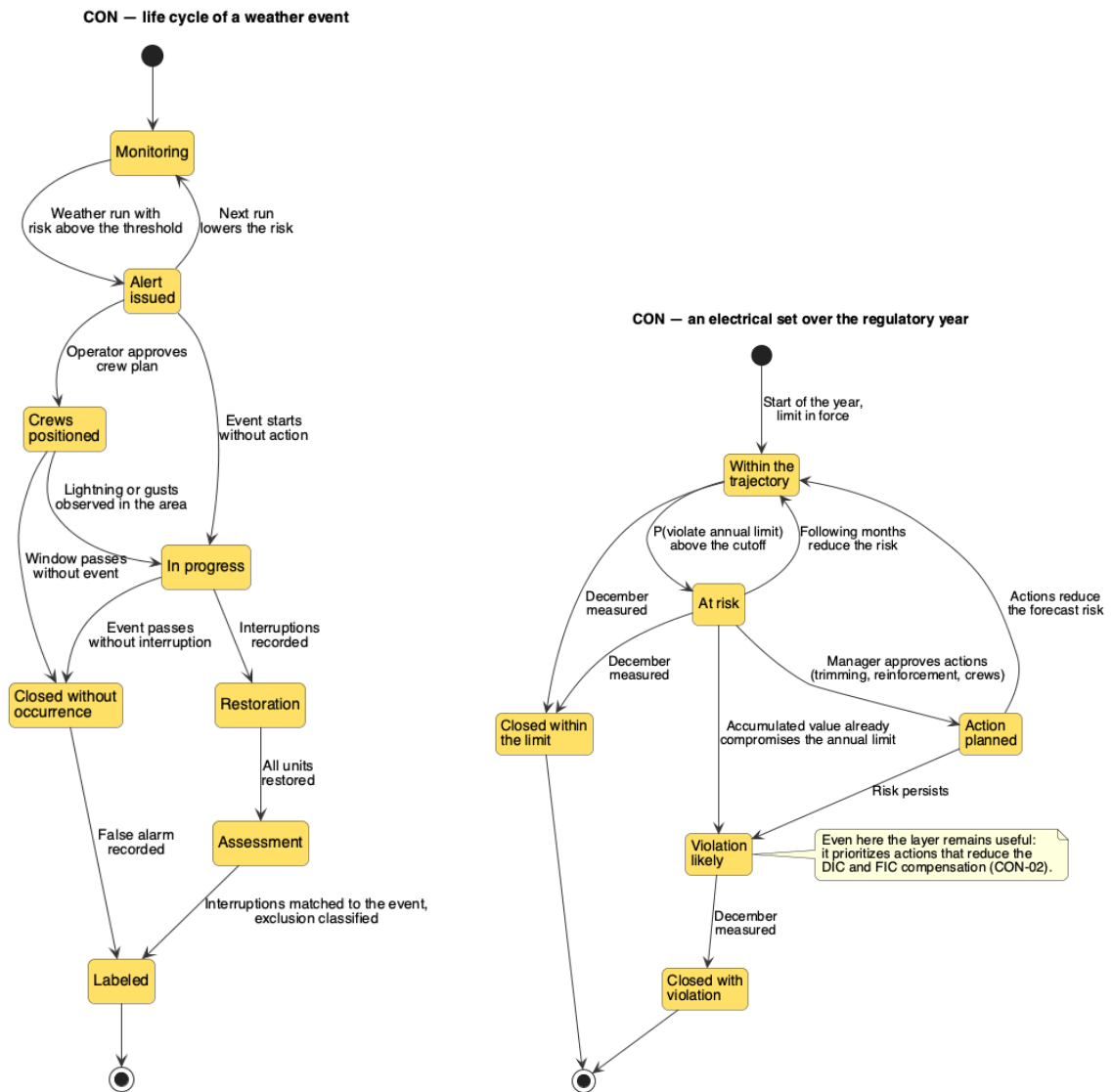


Fig. 4. CON: state machines of the weather event (left) and of the electrical set in the regulatory year (right). Labeled events, including false alarms, return as training labels.

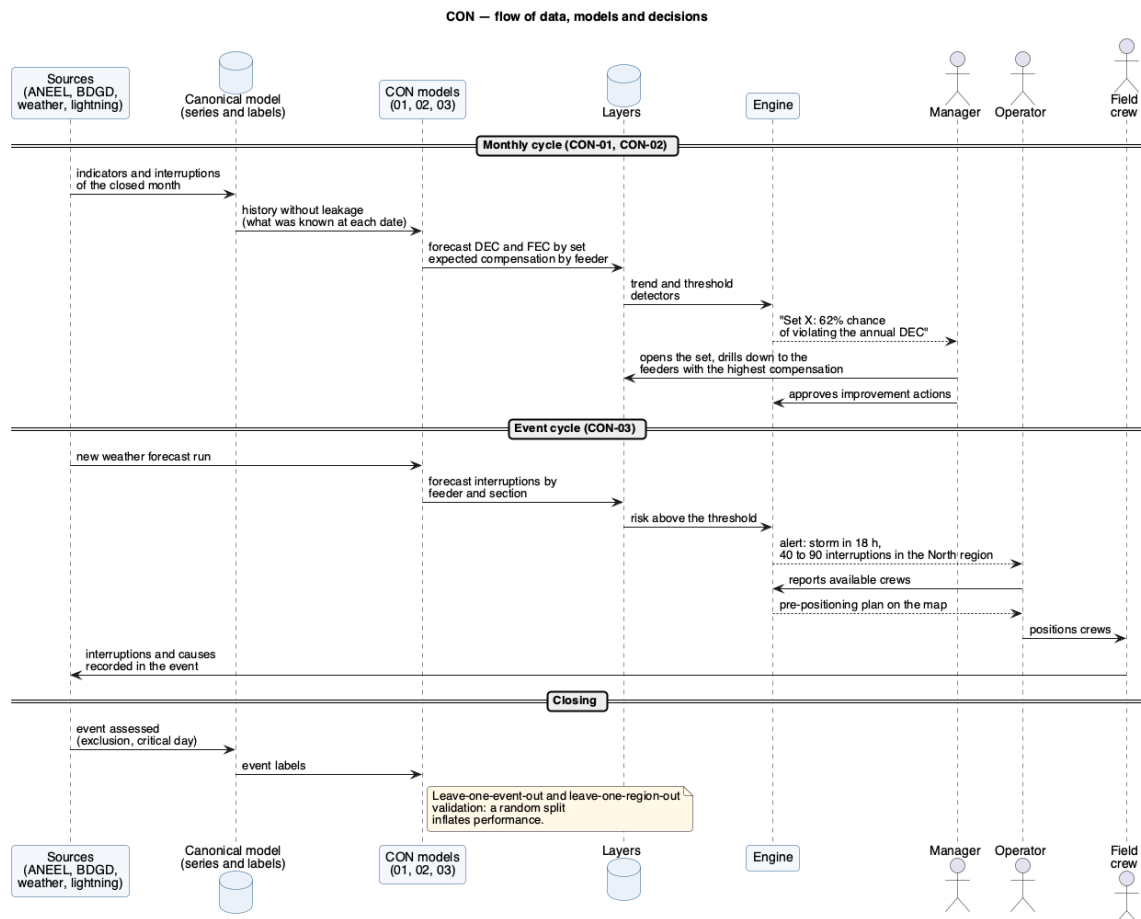


Fig. 5. CON: flow of the monthly cycle, the event cycle and the closing, with leave-one-event-out and leave-one-region-out validation.

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7 Assets and renewable generation

Shared sources

- **ONS Dados Abertos** (CC-BY, also mirrored on AWS) [8]: generation by plant on an hourly basis [5], hourly capacity factor of wind and solar plants [6], half-hourly constrained-off by plant since 2021 [1–4], availability of lines and transformers [7]. ONS warns that the data are revised after publication.
- **ANEEL**: SIGA (plant registry by CEG) [9], SIGEL (plants and lines in shapefile, wind turbines in SIGEL-EOL) [10], BDGD [11], interruptions in the distribution networks [12].
- **Join key**: the constrained-off data carry `ceg` and `id_ons` [4], which links ONS, SIGA and SIGEL through the CEG; plant groups appear without a CEG and require a dedicated correspondence table.

7.1 ATV · Assets and maintenance

ATV-01 · Power transformer health index

V3

Data. From the client: dissolved gas chromatography (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , CO , CO_2), oil physicochemical tests, furans, loading, age, work order history. Standards: IEC 60599:2022 [28]; IEEE C57.104-2019, which replaced total gas thresholds with percentiles by gas, deltas between samples and annual rates [27]; Duval triangles and pentagons [30]; CIGRE TB 761 assessment indices [31]. Public reference data: IEC TC10, with 117 labeled samples [29].

Approaches. Baseline: standard-based rules combined into a CIGRE-style weighted index. Recommended: gradient boosting that uses the standards as variables (ratios, Duval coordinates, deltas, rates, loss of life

by IEEE C57.91 [33]), with the later event (failure or intervention) as the label and not the Duval class; with calibration and SHAP. Recent review: Zahra et al. (2025) [32].

Metrics. F1 by fault class (diagnosis); AUC-PR for failure or intervention within 12 months and hit rate at the top of the list (prognosis); agreement with an expert.

Pitfalls. Validation on IEC TC10 alone (117 samples) does not support generalization; rates require three or more samples, rare in practice; a change of laboratory and oil treatment create artificial jumps.

ATV-02 · Failure risk by equipment

V3

Data. Public: BDGD (segments, transformers, switches, installation date) [11] and interruptions by event [12]; ONS for the basic grid [7]. From the client: registry with replacements, structured work orders (ATV-05).

Approaches. Baseline: Weibull by class and Cox. Recommended: random survival forest or boosting with a survival loss and time-varying covariates (weather, loading), handling censoring and left truncation [34,35]. Recent literature: DeepSurv and discrete-time neural models [36]. At the feeder level: failure count per km and year.

Metrics. C-index (including time-dependent), Integrated Brier Score, calibration at 12 months, hit rate at the top of the list.

Pitfalls. Left truncation (survivor bias); replacements without a record; external causes mixed with wear (competing risks).

ATV-03 · Vegetation near the network

V3

Data. Network geometry (BDGD [11], SIGEL [10]). Public canopy height: global 10 m map for 2020 (Lang et al. 2023) [37] and 1 m map (Meta and WRI, Tolan et al. 2024, images from 2017 to 2020) [38]. Sentinel-2 for updating. Client LiDAR when available.

Approaches. Baseline: canopy height and NDVI intersected with a buffer of the span. Recommended: dense vegetation segmentation and height regression with Sentinel-2 and GEDI; Brazilian case (SENAI and Cemig, 2025/2026) with RMSE of 7.7 m and MAE of 5 m against LiDAR [39]. With LiDAR: direct distance between conductor and vegetation. Recent literature: SAM adapted for transmission corridors [40].

Metrics. IoU and F1 of the segmentation; height error against LiDAR; recall of critical spans in the field; correlation with vegetation interruptions [12].

Pitfalls. An error of 5 to 8 m is of the order of the safety clearance zone: it serves to prioritize, not to measure distance. Static maps age with growth and tree trimming.

ATV-04 · Defects in inspection images

V4

Data. Client drone images. Public: CPLID (synthetic defects) [21]; InsPLAD, with 10,607 images, 17 asset classes and 6 defect types, but only 402 defect samples [22]; TPLA (towers and cables) [23]; UPID [24].

Approaches. Baseline: YOLO fine-tuned for components [26]. Recommended: two stages, detect the component and classify the defect or detect an anomaly in the crop [22]. Recent literature: RT-DETR (CVPR 2024) [25]; SAM for pre-annotation with a human in the loop.

Metrics. mAP by class; recall by critical defect; precision by structure (what becomes a work order).

Pitfalls. Rare defects; models trained on synthetic defects may not generalize; Brazilian insulator types differ; each image has to be associated with the BDGD structure.

ATV-05 · Structuring of work orders and causes

V3

Data. Free text of the client's work orders; causes from the interruptions dataset as a weak label [12]. Schemas: MaintIE (224 entities, 6 relations) [41] and ISO 14224 for failure modes [42].

Approaches. In failure mode classification into 22 ISO 14224 codes, a fine-tuned GPT-3.5 reached F1 0.80, against 0.60 for a text classifier and 0.46 for the model without fine-tuning [42]. Recommended: language model with JSON output constrained to a schema (component, failure mode, cause, action), client examples, light fine-tuning and human review by sampling.

Metrics. F1 by field; valid JSON rate; C-index gain in ATV-02 with the extracted variables.

Pitfalls. The benchmarks are in English and the work order text is short, abbreviated and regional; each extracted field has to keep the excerpt that justifies it; LGPD.

ATV-06 · Remaining useful life and maintenance plan

V4

Data. ATV-01 and ATV-02; client costs, crews and budget; consequence of failure by the affected customers (BDGD, interruptions).

Approaches. Baseline: risk $\times$ criticality matrix. Recommended: monetized risk by asset and integer optimization under crews and regions (to be confirmed as the standard). Recent literature: stochastic optimization under the uncertainty of the survival curve.

Pitfalls. Few end-of-life cases to validate; the optimization amplifies calibration errors.

7.2 GER · Renewable generation

GER-01 · Forecast generation by plant (P10–P90)

V2

Data. Location and specifications: SIGA and SIGEL [9,10]. **Public observed generation:** ONS by plant, hourly [5], and half-hourly verified generation in the constrained-off data [4]; covers Type I, II-B and II-C plants (II-C aggregated into groups). Weather forecast: ECMWF opened its whole real-time catalog under CC-BY in October 2025 [53]. Client SCADA in 10-minute averages.

Approaches. Baseline: persistence, climatology, power curve applied to the forecast. Recommended: quantile gradient boosting on numerical weather prediction variables, the winning method of GEF-Com2014 [52], with conformal recalibration. Onboarding without history: foundation models with the weather forecast as a future covariate (Chronos-2 [46], TimesFM 2.5 [47], Moirai 2.0 [48]); synthetic histories generated by a physical model reduced the error by 1.7 to 2 times at 440 photovoltaic plants [50]. In [49], no approach had the lowest error in all cases. In Brazil, ERA5 outperformed MERRA-2 for wind [54].

Metrics. nMAE and nRMSE by capacity; pinball and CRPS; P10–P90 coverage; skill against persistence; by horizon and without curtailment.

Pitfalls. **Train on the reference generation, not on the delivered generation**, so as not to learn the curtailment [4]; ONS revisions; leakage through observed weather; UTC versus local time.

GER-02 · Ramp alerts

V2

Approaches. There is no standard definition: a ramp is magnitude, duration (typically 1 to 4 h), time and direction, with a threshold by plant size [55]. Baseline: variation threshold on the P50. Recommended: ramp probability computed on **time-coherent scenarios**, not on marginal quantiles, which create spurious ramps.

Metrics. POD, FAR and CSI with a phase tolerance of one hour; Brier; lead time; alerts per week.

GER-03 · Satellite nowcasting (0 to 6 h)

V3

Data. GOES-19, operational GOES-East since April 4, 2025, on AWS without an account [13,14]; ABI with visible at 0.5 km and full disk every 10 minutes [15]; surface irradiance (DSR) at 2 km and 10 minutes in the Enterprise algorithm [16]; daily radiation from INPE [17].

Approaches. Baseline: persistence of the clear-sky index and cloud motion vectors. Recommended: U-Net on cloud index sequences, converted into power by plant [57,60]; Brazilian case with GOES-16 in Petrolina [59]. Recent literature, probabilistic: SHADECast, which extends the useful horizon by about 26 minutes [56]. Beyond 2 to 3 h, combine with numerical weather prediction [53].

Metrics. RMSE of irradiance and power; skill against persistence by lead time; CRPS.

Pitfalls. Parallax and resolution against small plants; latency; discontinuity in the switch from GOES-16 to GOES-19.

GER-04 · Curtailment detected and flagged

V2

Data. ONS constrained-off, dictionary v1.5 (September 2026): generation, limited generation, availability, reference, final reference (only for REL, sent to CCEE), reason (REL, CNF, ENE, PAR), origin (LOC, SIS), unrealized energy and minutes by reason [4]. Report of the ONS GT Curtailment [62]. SCADA with setpoints.

Approaches. Baseline: rules on the limited generation and the constraint minutes; in SCADA, setpoint below rated power. Recommended: **available power** model (normal behavior, GER-05); curtailment is available minus delivered when there is a limit, which detects unreported cuts and contests the ONS reference; power curve cleaning by anomaly detection [64].

Pitfalls. The 30-minute resolution hides short cuts; reimbursement rules change by resolution (to be confirmed).

GER-05 · Actual × expected performance

V3

Data. SCADA, resource, curtailment from GER-04. Public, for development: Kelmarsh and Penmanshiel (10-minute SCADA, CC-BY) [18,19]; PVDAQ from NREL [68].

Approaches. Baseline: power curve by bins and performance ratio. Recommended: normal behavior model (boosting or GAM) trained on a healthy period; for photovoltaic, fitted pvlib. Tools: OpenOA (NREL) [65] and RdTools for degradation [71].

Pitfalls. Training the expected value on data that already contain the loss; the cleaning may retain only about 31% of the data [64].

GER-06 · Turbine health

V3

Data. SCADA with bearing, generator and gearbox temperatures. Labeled: CARE to Compare (Fraunhofer, 2024), 95 datasets, 36 turbines, 44 labeled events, with the CARE metric (coverage, accuracy, reliability, earliness) [20].

Approaches. Review: Tautz-Weinert and Watson (2017) [66]. Chesterman et al. (2023), at 5 plants: Elastic Net competitive with LightGBM and neural networks; normalize by the fleet median; exclude the 4 months before the failure from training; no family dominates [67]. Recommended: normal behavior with fleet normalization and alarm with persistence.

Metrics. CARE score; days of lead time; false alarms per turbine per year.

Pitfalls. Few failures; pre-failure data contaminate the normal; climate of the Northeast different from that of the European datasets.

GER-07 · Soiling and solar cleaning schedule

V4

Approaches. Kimber (linear accumulation until rain above a threshold) and HSU (by particulate mass), both in pvlib [69,70]; from production, SRR (Deceglie et al. 2018) in RdTools [71]; report IEA PVPS T13-21:2022 [72]. Thresholds of rain that cleans range from 0.3 to 20 mm/day [73]. Decision: clean when the expected loss until the next forecast rain exceeds the cost.

Pitfalls. Confusing soiling with degradation or shading; unrecorded cleanings break the SRR.

GER-08 · Forecast aggregate generation by submarket

cond

Approaches. Sum of the forecasts by plant with upscaling for those without a forecast; hierarchical reconciliation (plant, state, submarket) [74] and a value-oriented version [75]. Aggregate quantiles require joint scenarios: the sum of quantiles is not the quantile of the sum.

Pitfalls. Entry of new plants (update through SIGA); stating whether the forecast is of available or delivered power; invisible DG.

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8 Market, weather, demand response and engine

8.1 MER · Market and price

NOTE Conditional track. The three features in this module, with CAR-08 and GER-08, are outside the phased roadmap: they enter only if the target client question is settled in favor of generator and energy trader. The research is recorded so that the decision, when it comes, does not start from zero.

NOTE Access to CCEE data. The CCEE Dados Abertos portal has a security block (application firewall) that stops clients outside the access policy and directs them to open a ticket with the error code and the IP [1,2]. The CCEE publishes tutorials on automation by API, and the download from a CKAN endpoint (/datastore/dump/<id>) worked without a block on September 23, 2026. Automated collection was possible with few requests, an identifiable user-agent, a local cache and download of the yearly CSVs instead of scraping.

MER-01 · Forecast PLD by submarket (quantiles)

cond

Data. *PLD_HORARIO* from the CCEE: one CSV per year, hourly from 2021 to 2026 and weekly from 2001 to 2020, CC-BY 4.0 license, daily update [3]. Regulatory limits for 2026: floor R\$ 57.31/MWh, structural cap R\$ 785.27/MWh and hourly cap R\$ 1,611.04/MWh (Despacho ANEEL 3.850/2025) [5]. NEWAVE/DECOMP/DESSEM decks in the Acervo CCEE, including shadow decks [6]; DESSEM decks also at the ONS [7]. ONS Dados Abertos: half-hourly load, generation by plant and half-hourly CMO [8].

Approaches. Baseline: seasonal naive and LEAR (autoregressive with LASSO), used as a reference and implemented in *epftoolbox* [10]. Recommended: gradient boosting with quantile loss on CMO, forecast load and renewables, reservoir storage and the deck in force, with conformal calibration (to be confirmed).

Recent literature: emulation of DESSEM by a surrogate model and networks that forecast parametric distributions [10].

Metrics. Pinball and CRPS, coverage and width of the intervals, MAE relative to the naive, Diebold–Mariano test.

Pitfalls. Structural break on January 1, 2021 (the PLD became hourly, computed by DESSEM) [4]; new future cost function on October 12, 2025 [6]; distribution censored at the floor and at the cap; ONS series revised after publication, which requires storing what was known at each date [8]. The Projeto Meta II proposes bid-based pricing and two PLDs (ex-ante and ex-post), with rollout planned for June 2028 [9], which will break the series if rolled out.

MER-02 · Joint scenarios of generation, load and price

cond

Data. Probabilistic outputs of GER-01, CAR-08 and MER-01; joint history from the ONS and the CCEE [3,8].

Approaches. Baseline: independent scenarios or bootstrap of historical blocks. Recommended: Gaussian copula on the predictive marginals (Pinson et al. 2009) [11]; alternative with the order of the members of the weather ensemble (ECC and dual-ECC) [12]. Recent literature: vine copulas and deep generators (to be confirmed).

Metrics. Energy score and variogram score, multivariate rank histograms, CRPS by marginal (*scoringrules* library) [13].

Pitfalls. The dependence changes with the hydrological regime, and a stationary copula underestimates extreme joint events (drought with peak load).

MER-03 · Bid optimization and battery dispatch

cond

Data. Scenarios from MER-02, hourly PLD and technical parameters of the battery. Context: Portaria MME 136/2026 defines the first capacity reserve auction for batteries (December 2026, 15-year contracts starting in August 2028), with availability revenue and not free arbitrage [14].

Approaches. Baseline: deterministic optimization on the point forecast. Recommended: two-stage stochastic programming or CVaR, on a rolling window. Recent literature: decision-focused learning; minimizing the CRPS does not guarantee the best bid [15].

Metrics. Revenue against the optimum with perfect information (regret), CVaR of revenue, degradation cost.

Pitfalls. Today the PLD comes from a cost model, not from bids; “bid” only makes sense with the hybrid model (2028 onward) [9]. Simplified degradation overestimates revenue. Low map fit.

8.2 CLI · Weather context

CLI-01 · Forecast weather

V2

Data. ECMWF Open Data: IFS and AIFS (deterministic and ensemble), 0.25° grid, four runs per day, CC-BY 4.0 license with commercial use allowed and attribution required; the portal keeps only the runs of the last 2 to 3 days, with cloud mirrors [16,17]. NOAA GFS 0.25° on AWS, open, with long history at NCAR [18]. CPTEC/INPE: BAM, WRF, Eta and BRAMS, with citation required [19]. INMET: hourly automatic stations (portal with 90 days) and BDMEP for the history, with a lag of 1 to 3 months [20].

Approaches. Baseline: raw forecast interpolated to the entities. Recommended: statistical post-processing by entity (MOS, EMOS, quantile regression) with downscaling. Recent literature: global ML models (AIFS, GraphCast-GFS) [16,18]. Tools: *herbie*, *xarray/cfgrib*, *xskillscore* [17].

Metrics. RMSE and MAE by variable and horizon, CRPS of the ensemble, reliability diagram, skill against climatology and persistence.

Pitfalls. **Archive every run from the first day:** without that, there is no history of forecasts to train without leakage. Changes of model version break the post-processing. The ECMWF attribution has to appear on the map.

CLI-02 · Observed satellite and radar

V2

Data. GOES-East has been **GOES-19** since April 2025 (75.2°W), with ABI and GLM on AWS Open Data in NetCDF4; there were outages (ABI restored in July 2026) [21]. REDEMET/DECEA radar by API with a key, but as a PNG image, useful to display and not to model [22]. CEMADEN: rain gauges and dual-polarization radars, with volumetric data on request (to be confirmed) [23].

Approaches. Baseline: display mosaics. Recommended: precipitation nowcasting from 0 to 6 h with *pysteps* (optical flow and stochastic ensembles) [24]. Recent literature: deep nowcasting (ConvGRU, diffusion).

Metrics. CSI, POD and FAR by threshold; FSS; CRPS [24].

Pitfalls. Uneven radar coverage across the territory; gaps and latency of GOES.

CLI-03 · Lightning and severe events

V3

Data. BrasilDAT (ELAT/INPE): detects cloud-to-ground and intracloud strokes; access by contacting ELAT, **it is not open** [25]. Open alternative: GLM on GOES-19, optical detection of total lightning with uniform efficiency over Brazil [21,26].

Approaches. Stroke density by cell and window; space-time join of lightning with feeder outages (feeds CON-03).

Metrics. POD and FAR of the alerts against outages; lead time.

Pitfalls. The GLM records optical flashes, not strike points, with location on the order of kilometers (to be confirmed): it does not serve to associate a stroke with a line span.

8.3 DRE · Demand response and consumer

DRE-01 · Demand response targets by feeder

V4

Data. Regulation: REN ANEEL 1.040/2022 (structural program) and the ONS sandbox for the availability product, with the first competitive mechanism in October 2024 (up to 4 monthly activations of 4 h, between 6 p.m. and 10 p.m.) [27,28]. Results at the CCEE [28]. The ONS program targets large consumers of the SIN; demand response by feeder is a distribution utility case and depends on its own metering (to be confirmed).

Approaches. Measurement baseline: “X of Y days” rules (to be confirmed). Recommended: counterfactual baseline by ML, evaluated as a counterfactual because it defines payment; recent literature: generalized synthetic control and probabilistic baselines [29]. Targets: feeders with forecast loading above the limit (CAR-01, RED-01), ranked by expected relief per cost.

Metrics. Bias of the baseline on days without an event; precision at top-k of the chosen feeders; MW reduced against contracted.

Pitfalls. A biased baseline pays unduly or gives an incentive to inflate the load before the event; rebound effect; few events to validate.

DRE-02 · Consumption disaggregation (NILM)

out

Data. High-frequency aggregate metering by unit; public datasets REDD and UK-DALE; no Brazilian dataset found [30].

Approaches. FHMM (NILMTK) as baseline; seq2point (Zhang et al. 2018) as recommended [30].

Why it is out. It requires high-frequency individual data that does not exist at scale, the Brazilian appliance stock (electric shower) differs from the training datasets, and the output is by consumer unit, not spatial.

DRE-03 · Tariff and efficiency recommendation

out

Data. Tarifa Branca since 2018, uptake of about 69 thousand units out of 75 million eligible [31,32]; proposal by ANEEL’s technical staff (November 2025) of automatic Tarifa Branca for consumption above 1 MWh/month, taken to public consultation, outcome not verified [32]. Approved tariffs by distribution utility in ANEEL’s open data [33].

Why it is out. Without a measured hourly curve the recommendation is speculative, and the decision

belongs to the consumer unit, not to an entity on the map. If the automatic Tarifa Branca is approved, the aggregated version (how many units of a feeder would be affected) can return as an insight.

8.4 ENG · Analysis engine and models

ENG-01 · Consolidated insight feed

MVP

Data. All layers with the hierarchy, a neighborhood matrix (geographic or by electrical topology) and the history by cycle.

Approaches. *Scoring:* adopt the scheme of Top-K Insights (SIGMOD 2017) and QuickInsights (SIGMOD 2019, basis of the Power BI feature): insight type, subspace and score by impact $\times$ significance [34]. *Spatial anomaly:* local Moran (LISA) and Getis-Ord G_i^* in PySAL/esda, with pseudo-p by permutation [35], always with FDR control for multiple and dependent tests [36]. *Change and trend:* offline change points with *ruptures* and online with BOCPD [37]; exceedance of a forecast quantile for thresholds. *Consolidation:* the parent's insight absorbs those of the children when it explains most of the effect; MinT reconciliation guarantees coherent numbers across levels [38].

Metrics. Precision at top-k of the feed (insights marked useful or acted on), false positive rate under shuffled data, insights per real event.

Pitfalls. Explosion of tests (entities $\times$ layers $\times$ detectors); MAUP (the result changes with the aggregation scale); geographic neighborhood is not electrical connectivity; alert fatigue.

ENG-02 · Explanation by entity

MVP

Approaches. TreeSHAP for tree models [39]; TimeSHAP for time series models [40]; GeoShapley for spatial models, which treats location as a joint player [41].

Metrics. Fidelity (deletion and insertion), stability across runs, consistency with the physics (expected sign of temperature on load).

Pitfalls. Correlated weather variables give different attributions depending on the SHAP variant [39]; explanation is not cause and must not feed the scenario simulation as if it were effect.

ENG-03 · Model error by entity

MVP

Data. Forecast $\times$ actual pairs, versioned by issue date, horizon and model version; actuals revised by the ONS and the CCEE require storing the version known at each date [8].

Metrics. nMAE with explicit normalization; CRPS (strictly proper rule) [42]; pinball; calibration (PIT, coverage, reliability) and sharpness; precision at top-k for rankings. Libraries: *scoringrules* (active, univariate and multivariate) and *xskillscore*; *properscoring* is poorly maintained [42].

Pitfalls. nMAE blows up in low-load entities; aggregate metrics hide the tail; few samples per entity call for a bootstrap interval before issuing the performance insight.

ENG-04 · Data drift by region

V2

Approaches. Univariate tests (KS, PSI, Wasserstein) as in *Evidently* [43]; MMD for multivariate drift; performance estimation without labels with NannyML [45]; two-sample tests on reduced representations work better according to Rabanser et al. (2019) [46]. Apply the test by region and look for drift clusters with LISA (proposal, to be confirmed as practice).

Pitfalls. **License:** *alibi-detect* changed from Apache 2.0 to Business Source License 1.1 in January 2024, with a restriction on commercial use in production [44]. With large samples, small differences become statistically significant: use effect size and FDR; compare with the same season of the year.

ENG-05 · Scenario simulation

V2

Approaches. Perturb inputs and rerun the models, changing correlated variables coherently and flagging extrapolation outside the training domain. Where there are joint scenarios (MER-02, on the conditional track), propagate the uncertainty over them; without them, perturb the marginals and state that the dependence is not modeled. Recent literature: structural causal models (to be confirmed: DoWhy, EconML).

Metrics. Backtest on known historical shocks; coverage of the intervals.

Pitfalls. Predictive models are not causal; the user may read the scenario as a forecast.

ENG-06 · Free-text question

V2

State of practice. In text-to-SQL on real databases (BIRD), the best system reported in 2025 reaches about 82% execution accuracy against 93% for humans [47]. In geospatial tasks, agents frequently “hallucinate” spatial relations instead of computing them [48]. In DataGemma, even with data retrieval, a relevant share of the statistics cited came out wrong [49]: the language model cannot be a source of numbers.

Recommended architecture. (1) the model chooses tools from a restricted semantic API (metrics, entities, filters); free SQL only as an exception resource, read-only and with limits; (2) the engine executes and returns numbers with identifiers; (3) the text is written with placeholders filled by the engine; (4) a verifier rejects the answer if any number does not come from the result; (5) the answer cites the insights and queries and admits “I don’t know”.

Metrics. Execution accuracy on an in-house set of questions from the sector; rate of numbers without a source (target: zero); correct refusal rate.

Pitfalls. Wrong unit (MW and MWh); repeated substation names; instruction injection coming from text in the data; access to data of consumer units (role-based control in the tool layer).

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